

What the September scrape actually contains

A programmatic audit of the three raw workbooks behind Shelfline — before any ingestion pipeline, schema, or API gets built on top of them. No data was modified, moved, or renamed.

Content Data

88,307 rows 137 cols

55.3 MB

5 snapshots · weekly (Thu)

Price Data

529,803 rows 67 cols

148.5 MB

30 snapshots · daily, no gaps

Share of Search

237,064 rows 235 cols

258.1 MB

4 snapshots · weekly (Tue)

A Dataset overview

Three standalone Excel workbooks live under `data/September/`, one per data domain, each a single sheet with a header row and no partitioning by day. Combined: **461.9 MB**, **855,174** data rows, 439 raw columns across the three files. There is no archive/zip wrapper — each `.xlsx` is the terminal artifact.

FILE	SHEET NAME	ROWS	COLS	SIZE	DATE RANGE	CADENCE
Content Data	20220901-20220930_Content_Data	88,307	137	55.3 MB	Sep 1 – 29	5 dates, weekly
Price Data	20220901-20220930_Price_Data	529,803	67	148.5 MB	Sep 1 – 30	30 dates, daily
Share of Search	010922_300922_Share of Search	237,064	235	258.1 MB	Sep 6 – 27	4 dates, weekly

Rows/day are near-constant within each file (Content $\approx 17.6K \times 5$, Price $\approx 17.6K \times 30$, Share of Search $59,266 \times 4$) — each crawl re-observes essentially the same fixed universe rather than a rotating sample, which is good news for trend reliability.

FIRST FINDING THAT CHANGES EVERYTHING DOWNSTREAM

The three files use **three different, non-overlapping date sets**: Content on the 1st/8th/15th/22nd/29th, Price daily, Share of Search on the 6th/13th/20th/27th. No exact calendar date is shared by all three. Any cross-file join must key on **ISO week**, not on `crawl_date` equality — see Section E.

B Schema

Representative columns by domain, with fill-rate and cardinality from a full streaming pass (not a sample). Full 137/67/235-column manifests were generated as JSON alongside this report for engineering handoff.

Content Data — key columns

COLUMN	TYPE	NON-NULL	UNIQUE	CATEGORY
Crawl date	date	100%	5	Time
Retailer site	string	100%	62	Entity – retailer
Retailer id	string/int	99.95%	12,368	Identifier
Product name	string	89.07%	14,147	Entity – product
Brand	string	88.05%	241	Entity – brand
Category name	string (code)	100%	8	Entity – category
Upc	string	20.45%	2,374	Identifier
Asin	string	88.88%	10,893	Identifier (caveat – see C)
Stock status	string (free text)	89.07%	373	Commerce
Rating	float	78.73%	706	Reviews
Total reviews	int	78.73%	5,832	Reviews
Title / Product description	string	89.07% / 73.74%	high	Content
No of images / No of videos	int	89.04% / 17.33%	25 / 11	Content
Bullet 1...27	string ×27	77.1% → 0%	taper	Content (repeating group)
Varient label/value 1...22	string ×44	35.4% → 0.07%	taper	Content (repeating group)
Rank 1...4 / Category 1...4	int / string	42.3% → 0.65%	taper	Digital shelf (see G)
SPB Url / Front image	URL	100% / 89.04%	high	URL / identifier

Price Data — key columns

COLUMN	TYPE	NON-NULL	UNIQUE	CATEGORY
Crawl date	date	100%	30	Time
Retailer site / id	string / string	100% / 99.98%	63 / 12,090	Entity / identifier
List everyday price	mixed numeric/text	70.09%	high	Commerce (needs parsing – see D)
Current price	mixed numeric/text	23.59%	6,685	Commerce – populated only when discounted
Subscription price	mixed numeric/text	19.97%	5,608	Commerce

COLUMN	TYPE	NON-NULL	UNIQUE	CATEGORY
Coupon value	string ("13.44 (15%) ")	10.36%	3,187	Commerce (compound field)
Stock status	string (free text)	89.02%	545	Commerce
Buy box seller / shipper	string	74.27%	2,301 / 1,736	Digital shelf
Total reviews	int	78.92%	12,286	Reviews (daily-cadence version)
Site upc	string	18.89%	1,975	Identifier
Seller 1...15 / name / price tag	float / string / string	29.1% → 0.02%	taper	Competitor (marketplace offers, repeating group)

Share of Search — key columns

COLUMN	TYPE	NON-NULL	UNIQUE	CATEGORY
Crawl_date	date	100%	4	Time
site	string	100%	23	Entity – retailer
keyword	string	100%	5,293	Entity – search term
Url_1...76	URL ×76	84.8% → <0.01%	taper	Digital shelf (repeating group, rank slot)
Url_1...76_Sponsored	Yes/No	mirrors Url_N	3	Digital shelf
Product_name_1...76	string	mirrors Url_N	taper	Content (search-result label)

WHY THE REPEATING GROUPS MATTER

Bullets, variants, sellers, and rank slots are stored *wide* (one column per position, up to 76 in Share of Search). Fill rate collapses fast past the first few positions. These should become normalized child tables during ETL (Section I), not 27/44/45/76 individual Postgres columns.

c Entities

Retailers

62 (Content), 63 (Price), 23 (Share of Search — a US/Canada/LatAm subset). Amazon dominates by volume across every file (24.8% of Content rows, 24.8% of Price rows), followed by Chewy, Walmart, and the six Amazon country subsidiaries (.de/es/fr/it/pl/se). Geographic footprint spans US big-box (Walmart, Target, Home Depot, Lowe's, Menards), pet specialty (Chewy, Petco incl. Mexico, PetSmart), Kroger's regional banners (Kroger, Ralphs, QFC, Fred Meyer, Dillons, Gerbes, King Soopers), and LatAm/EU marketplaces (Mercado Libre ×3, Alkosto, Exito, Falabella, Coppel, Liverpool).

RETAILER NAMING IS NOT CONSISTENT ACROSS FILES

Acehardware.com (Content) vs. AceHardware.com (Price); meijer.com (Share of Search) vs. Meijer.com (Content/Price). A canonical retailer dictionary keyed on domain (case-folded) is required before any cross-file join or retailer-level rollup.

Brands & categories

241 raw brand strings in Content, 276 in Price — but many are casing/punctuation duplicates of one brand: IAMS / Iams , Marineland / MarineLand , Furminator / FURminator , Remington / REMINGTON , and at least four spellings of Black+Decker (BLACK+DECKER , Black & Decker , Black + Decker , Black+Decker). True canonical brand count is likely 150–180, pending an alias table. Category is cleaner in concept but still fragmented: only 8 raw values exist, and they collapse to 4 canonical business units — GPC , PET , HPC , HG — with HOME & GARDEN / H&G / Home as spelled-out aliases of HG , and Pet as an alias of PET . There is no granular product taxonomy (e.g. "Insect Repellents," "Slow Cookers") consistently populated beyond this 4-way split — the free-text Site category breadcrumb (86.6% filled, 2,348 distinct values) is the only granular signal, and it is retailer-specific, not a shared taxonomy.

Product identity — the load-bearing question

Tested four candidate identifiers plus one composite key on Content Data (the only file with enough repeat snapshots this pass to test persistence directly):

CANDIDATE KEY	FILL RATE	UNIQUE VALUES	STABILITY FINDING	VERDICT
Product name (alone)	89.1%	14,147	14,923 (site, name) pairs; 90.5% seen in all 5/5 weekly snapshots	● Usable fallback
Retailer id (alone)	99.95%	12,368	11.9% of ids collide across >1 unrelated retailer site	● Not unique alone
Retailer site + Retailer id	99.95%	17,702 pairs	98.8% (17,492) seen in all 5/5 weekly snapshots	● Best canonical key
UPC	20.45%	2,374	26.1% of populated UPCs span >1 retailer – genuinely cross-retailer	● Secondary linking signal
Asin	88.88%	10,893	Mirrors Retailer id on non-Amazon rows – likely mislabeled, not a real ASIN off-Amazon	● Trust only on Amazon.*

Recommendation: use (retailer_site, retailer_id) as the canonical product-per-retailer key. Use UPC, where present, as a secondary signal for cross-retailer "same physical product" matching — it cannot be a primary key at 20% fill.

D Data quality

Duplicates (nothing deleted – counts only)

FILE	KEY TESTED	KEYS	DUPLICATED KEYS	MAX REPEATS	ROWS AFFECTED
Content	retailer + product name + date	71,079	4,118	37	11,694 (13.2%)
Price	retailer + product name + date	425,972	24,756	37	70,433 (13.3%)
Price	UPC + date (cross-retailer, expected)	57,156	14,394	77	–
Share of Search	site + keyword + date	237,064	0	1	0%

The ~13% duplication rate in Content/Price on (retailer + product name + date) is consistent across both files — likely the same product being crawled under more than one site-category placement per snapshot, not random corruption. Share of Search is perfectly clean at the row-grain level. Note: grouping Price by `retailer_id + date` alone (omitting `retailer_site`) produces up to 216 "duplicates" for one key — that's an artifact of `retailer_id` not being globally unique (Section C), not a real data defect.

Missing data, prioritized

Critical (blocks identification or joins): UPC missing 79.6% (Content) / 81.1% (Price) — the only strong cross-retailer identifier is the sparsest field in the dataset. **Important** (blocks core analytics): Rating/Total reviews ~21% null; Current price 76.4% null (though this is largely structural – see below); Stock status ~11% null. **Optional**: Ingredients list (80.7% null), Image 360 (99% null), No of videos (82.7% null), variant fields past position 5.

About 10.9% of Content rows (9,652) are essentially crawl-failure placeholders — Title, Description, Bullets, Rating, and Stock status are *all* null together on these rows, correlated with a non-null `url_fail` /redirect flag. This is a structural "crawl didn't resolve" pattern, not scattered missingness, and should be modeled as a distinct observation status rather than 8 separate null checks.

"Current price" at 23.6% fill is **not** a data-quality problem — sampling shows it is populated specifically when an item is discounted vs. "List everyday price" (70.1% fill, the base price). Treat effective price as `COALESCE(current_price, list_everyday_price)`.

Anomalies

- Price fields are mixed-type.** Of the populated "List everyday price" cells, 110,354 (~30%) are non-numeric strings, not floats — likely currency-formatted or locale-specific text mixed in with clean numbers. "Subscription price" is worse: 44.6% non-numeric. Every price column needs a defensive parser before any arithmetic.
- Coupon value is a compound field:** values like `"13.44 (15%)\n"` pack an absolute discount and a percentage into one string — needs regex extraction into two numeric columns.

- **No negative prices, no negative review counts, no out-of-range ratings** (min 1 / max 5, exactly as expected) — the numeric core of the data is clean where it is numeric.
- **Extreme-but-plausible prices:** 1,970 "List everyday price" rows exceed \$5,000 (max \$57,999); 60 "Current price" rows do (max \$55,099). Consistent with appliances/high-end SKUs in this catalog (mixers, irons, aquariums) — flag for spot-check, don't blanket-reject.
- **Stock status is a free-text mess:** 373 distinct values in Content, 545 in Price, across ~7 languages, mixing true status ("In Stock.", "Out of Stock", "Auf Lager.", "Dostępny.") with quantity messages ("Only 1 left in stock – order soon.") and leaked CTA button text ("Agregar a carrito", "Comprar ahora"). Needs a keyword/regex-driven mapping to a small canonical enum.

E Time-series coverage

FILE	SNAPSHOTS	DATES (SEPT 2022)
Content	5	1, 8, 15, 22, 29 (Thursdays)
Price	30	1–30, every day, zero gaps
Share of Search	4	6, 13, 20, 27 (Tuesdays)

No calendar date is shared across all three files. Content and Share of Search are both weekly but on different weekdays and different phase — they align only at **ISO-week** granularity (e.g. "week of Sep 5" pairs Content's Sep 8 crawl with Share of Search's Sep 6 crawl). Price's daily cadence can be downsampled to whichever week-anchor is chosen, or kept daily for price-trend-only views.

Product persistence (Content, tested directly)

Of 17,702 unique (retailer_site, retailer_id) product identities seen in September, **98.8% (17,492)** appear in all 5 of 5 weekly snapshots — very low catalog churn at weekly grain, which means trend and availability analytics on this key will be reliable. The weaker (retailer_site, product_name) key persists at 90.5% — string-based matching loses ~8 points to naming drift/typos. Price Data's near-constant ~17.6K rows/day across all 30 days is consistent with the same fixed universe being tracked daily, though that persistence claim was not separately re-verified row-by-row in this pass and should be confirmed before relying on it for daily deltas.

Share of Search: rank depth is not uniform

15.2% of rows (36,092 of 237,064) returned **zero** ranked URLs for their keyword/site/date — worth confirming whether these are genuine no-result searches or crawl failures before trusting visibility metrics at face value. Among rows that did return results, depth is strongly bimodal with spikes at 24, 36, 44, 48, 50, and 60 slots — consistent with different retailers paginating search results to different fixed page sizes, not random noise. Any "share of search" calculation must normalize by each site's typical result depth rather than assume a uniform top-N across retailers.

F Metrics supported

Classified against what the raw fields can actually deliver — not what would be nice to have.

METRIC	CLASSIFICATION	BASIS
Availability / stock status	● Direct	Stock status field, once normalized to a canonical enum (Section D)
Price (current, effective)	● Direct	<code>COALESCE(current_price, list_price)</code> , once parsed to numeric
Price change	● Derived	Day-over-day delta on daily Price Data (30/30 days available)
Price index (vs. category/competitor set)	● Assumption-based	Needs a documented "comparison set" definition (same category code + date, or same UPC across retailers)
Rating / review count	● Direct	Present in both Content (weekly) and Price (daily) files
Review count change	● Derived	Delta between snapshots — a volume trend, not sentiment
Content completeness	● Derived	Composite score from title/description/bullets/images/video presence – see Section H below for framework
Search rank / visibility	● Direct	Within the 5,293 tracked keywords only — not category-wide organic rank
Rank movement	● Derived	Week-over-week rank delta per (site, keyword, product)
Buy-box proxy	● Direct	Buy box seller/shipper fields, present in Content & Price
Retailer / category comparison	● Direct	Group-by on Retailer site / 4-way category code
Competitor listing comparison	● Derived	Via marketplace Seller 1–15 offers (Price) and co-occurring brands/products for a keyword (Share of Search) — not a true competitor master entity (see H)
Product opportunity score	● Assumption-based	Composite of the above; requires a documented weighting methodology, sign-off before shipping to UI
Estimated sales	● Not supported	See Section G
Market share	● Not supported	See Section G

Content-score inputs available

Title (89.1% fill), description (73.7%), bullet count (77.3%, up to 27 bullets), image count (89.0%), video presence (17.3%), variant/size-color options (35.4% have ≥ 1 variant). Sufficient for a simple, defensible weighted checklist score — not sufficient for anything claiming SEO or conversion impact.

Review analytics ceiling

Only aggregate `rating` and `total reviews` exist — no review text, no per-review date, no rating distribution, no sentiment field anywhere in either file. Trend-of-count and trend-of-average-rating are supportable; anything claiming sentiment or theme extraction is not, without bringing in an external review-text source.

G Sales & share feasibility

EXPLICIT FINDING

Across all 137 + 67 + 235 raw columns inspected, **there is no units-sold, revenue, transaction, traffic, conversion, or market-share field anywhere in this dataset**. Rating and review count are the only demand-adjacent signals, and both are cumulative, not incremental.

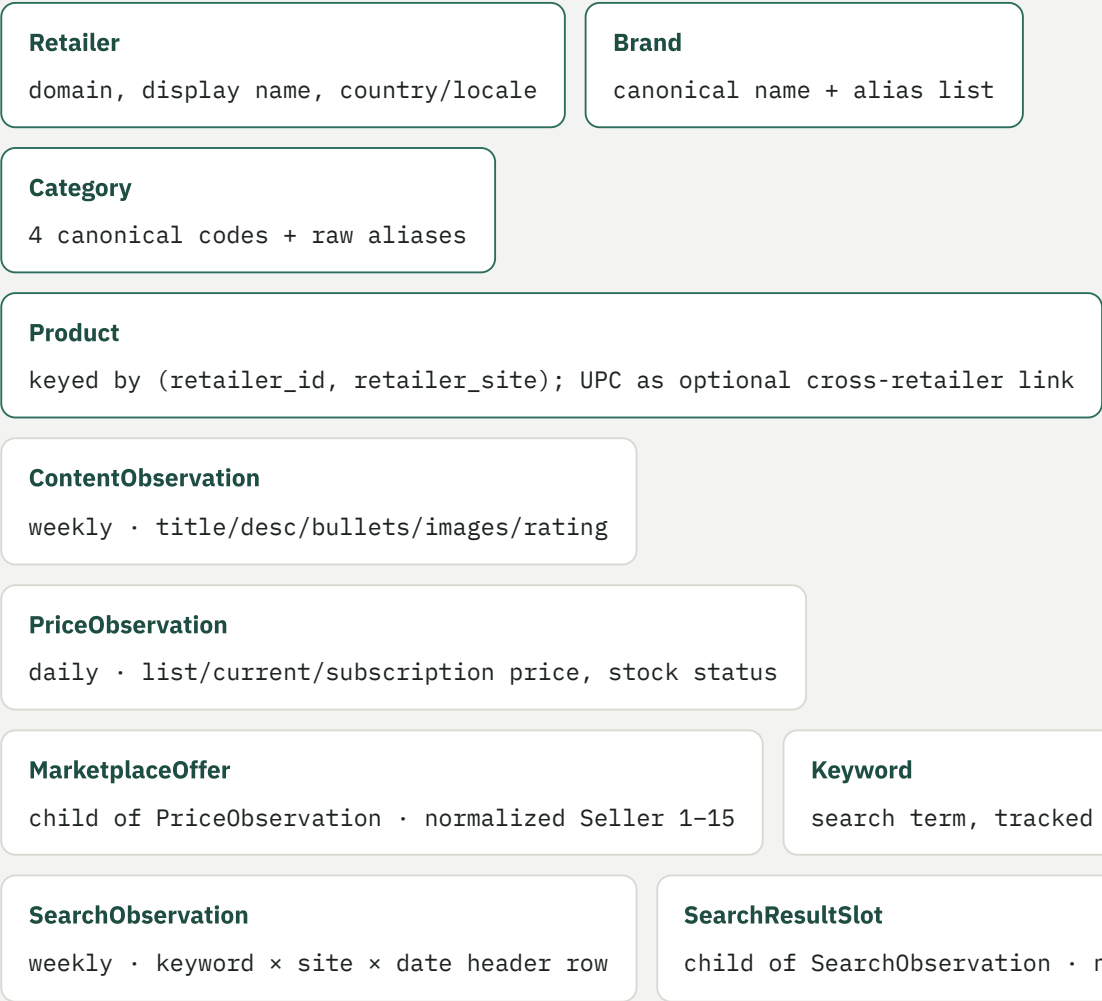
One field deserves a specific note: Content Data's `Rank 1-4 / Category 1-4` columns (42.3% filled at position 1, tapering to 0.65% at position 4) resemble Amazon's Best-Sellers-Rank pattern (large ordinal rank + category label, e.g. `#606,671 in Home`). **This is an unconfirmed hypothesis**, not a validated field — it needs to be checked against which `retailer_site` each populated row belongs to before being used for anything. Even if confirmed, BSR is an ordinal, Amazon-only, indirect sales proxy — not a units or revenue figure.

SHELFLINE METRIC	FEASIBILITY	NOTES
Estimated Sales	● D – Not supported	No sales data exists. A BSR-based estimate is possible only for Amazon rows, only if the Rank field is confirmed as BSR, and only via a documented, clearly-labeled estimation model with wide error bars.
Market Share (revenue/unit)	● D – Not supported	No revenue or unit base exists to compute share against.
Share of Search (search-visibility share)	● A – Directly supported	Computable today from the Share of Search file — but it is a share of <i>search-result placements for tracked keywords</i> , and must never be relabeled "market share."
Sales Growth	● D – Not supported	No sales base metric to grow.
Category Growth	● D – Not supported (as revenue)	A "category price/availability/visibility trend" is derivable and useful — but it is not category revenue growth and should not be presented as such.

Bottom line for product/UI: do not ship "Estimated Sales," "Market Share," "Sales Growth," or "Category Growth" as currently labeled. "Share of Search" can ship as-is with accurate labeling.

H Recommended canonical data model

Minimum entity set the data actually justifies — no Competitor or Sales entity, because nothing in the raw data resolves to either as a first-class object today.



Deliberately excluded: a first-class `Competitor` entity (nothing in the data names "who competes with whom" — it must be derived at query time from shared category + co-occurring brand/keyword), and a `Sale / Transaction` entity (no such data exists — see Section G).

I Raw → curated pipeline

STAGE	WHAT HAPPENS
Raw	Land the three <code>.xlsx</code> files untouched in object storage; never edit in place.
Validation	Schema check (column count/order per file), row-count sanity vs. prior load, reject-and-quarantine rows that fail type coercion instead of silently dropping them.
Normalization	Retailer/brand/category alias tables; stock-status enum mapper; price-string parser (currency symbols, locale decimal formats); coupon-value regex split; unpivot Bullet/Variant/Seller/Url_N repeating groups into long child tables.

STAGE	WHAT HAPPENS
Canonical entities	Resolve Product on (retailer_site, retailer_id); backfill UPC-based cross-retailer links where present; dedupe on (retailer, product, date) with a documented tie-break rule (keep latest crawl_date_time).
Observations	Write ContentObservation (weekly), PriceObservation (daily), SearchObservation (weekly) as append-only fact tables keyed by ISO week where cross-file alignment is needed.
Analytics	Pre-aggregate retailer/category rollups, price-change deltas, content-completeness scores, rank-movement deltas.
API	FastAPI reads from analytics tables, not raw observations, for latency.
React	Shelfline UI consumes the API — see Section L for what must change in what it currently shows.

J Recommended PostgreSQL schema

TABLE	PURPOSE	PRIMARY KEY	KEY COLUMNS	INDEXES
retailers	Canonical retailer dictionary	retailer_id (surrogate)	domain (unique, case-folded), display_name, country	unique(domain)
brands	Canonical brand + aliases	brand_id	canonical_name, alias[]	gin(alias)
categories	4 canonical business-unit codes	category_code	canonical_code, raw_aliases[]	—
products	One row per (retailer, site product id)	product_id (surrogate)	retailer_id FK, site_product_id, upc, brand_id FK, category_code FK	unique(retailer_id, site_product_id); btree(upc)
content_observations	Weekly content snapshot	(product_id, crawl_date)	title, description, bullet_count, image_count, rating, total_reviews	btree(crawl_date)
price_observations	Daily price/stock snapshot	(product_id, crawl_date)	list_price, current_price, stock_status_norm, buy_box_seller	btree(crawl_date); btree(product_id, crawl_date)
marketplace_offers	Normalized Seller 1–15	surrogate id	price_observation_id FK, seller_name, seller_price, condition_tag, offer_rank	btree(price_observation_id)
keywords	Tracked search terms	keyword_id	keyword_text, locale	unique(keyword_text, locale)
search_observations	Weekly keyword×site header	(keyword_id, retailer_id, crawl_date)	result_depth	btree(crawl_date)

TABLE	PURPOSE	PRIMARY KEY	KEY COLUMNS	INDEXES
search_result_slots	Normalized Url_1...76	surrogate id	search_observation_id FK, position, url, is_sponsored, product_name, product_id FK (nullable)	btree(search_observati on_id, position)

Partition `price_observations` by month (or `crawl_date` range) once volume grows past a single September; it is already the largest table by row count and grows daily. Everything else stays unpartitioned at this scale.

K Recommended API endpoints

ENDPOINT	DATA SOURCE	NOTES
GET <code>/v1/overview</code>	Rollups across content/price/search_observations	Retailer & category counts, latest crawl dates, coverage freshness – no sales figures
GET <code>/v1/digital-shelf</code>	price_observations + search_observations	Availability rate, price index, search visibility – filterable by retailer/category/brand/date
GET <code>/v1/digital-shelf/products</code>	products + latest price_observations + content_observations	Paginated product list with price, stock, content-completeness score
GET <code>/v1/digital-shelf/keywords</code>	search_observations + search_result_slots	Rank & rank-movement per keyword/site, sponsored vs. organic split
GET <code>/v1/content</code>	content_observations	Content-completeness score + underlying field checklist per product
GET <code>/v1/reviews</code>	content_observations / price_observations	Rating & review-count trend only – no sentiment, no verbatim text
GET <code>/v1/competitors</code>	marketplace_offers + brand co- occurrence in search_result_slots	"Competitive listing comparison," explicitly not a share-of-market view
GET <code>/v1/products/{id}</code>	All observation tables joined on product_id	Full product detail & history

L UI gap analysis

What the current Shelfline UI shows vs. what this dataset can honestly back up.

CURRENT UI METRIC	DATA SUPPORT	ACTION
Availability	• Direct	Keep — after stock-status normalization
Price / price change	• Direct	Keep — after price-string parsing fix
Search rank	• Direct	Keep, scope caveat: tracked-keyword set only, not full organic rank

CURRENT UI METRIC	DATA SUPPORT	ACTION
Content Score	• Derived	Define & publish the scoring methodology before displaying a single number
Rating / Reviews	• Direct	Keep; relabel as "review volume trend," not sentiment
Estimated Sales	• Not supported	Remove, or replace with a clearly-labeled, Amazon-only, caveat-heavy BSR proxy once confirmed
Market Share	• Not supported	Remove. Rename existing capability to "Share of Search" only
Sales Growth / Category Growth	• Not supported	Remove; replace with "category price/availability trend" if useful
Competitor metrics	• Derived	Modify to "Competitive Listing Comparison" (buy-box + marketplace offers + keyword co-occurrence) — not a share-of-market claim

M Recommended next steps

1. Confirm the Rank 1–4/Category 1–4 “BSR hypothesis” with the data provider, and check whether it's populated only on Amazon.* rows.
2. Build retailer-domain and brand alias-normalization tables from the raw value lists already extracted in this audit.
3. Build a defensive price-string parser (handles currency symbols, locale decimal formats, the compound coupon field) and unit-test it against the ~30–45% non-numeric cells found in each price column.
4. Build the stock-status canonical mapper (keyword/regex-based, locale-aware) targeting a 5–6-value enum.
5. Write the ETL step that unpivots Bullet/Variant/Seller/Url_N repeating groups into the normalized child tables from Section J.
6. Stand up a DuckDB/Parquet staging layer for fast local iteration before committing to Postgres DDL — these files are too large for comfortable ad hoc pandas work.
7. Implement entity resolution on (retailer_site, retailer_id) as primary key, UPC as secondary cross-retailer link, with a documented tie-break rule for the ~13% duplicate rows found per snapshot.
8. Draft and sign off on the content-completeness scoring methodology with stakeholders before it appears in any UI.
9. Circulate Section G (sales & market-share limitations) to product/design before any further UI work — this determines what gets removed vs. relabeled.
10. Build the Postgres schema and FastAPI layer per Sections J/K once 1–7 are in place.
11. Instrument a freshness/cadence monitor going forward, since the three sources already run on three different schedules — a future gap will otherwise look identical to a genuine missing week.

Methodology: full streaming pass (openpyxl, read-only mode) over all rows of all three workbooks – no sampling, no data modification. Column-level uniqueness capped at 15,000 distinct values per column for memory safety; affected columns are marked accordingly in the full JSON manifests. Source files untouched at `data/September/*`.